114-2 數位電子學
第二次小考檢討

2026/6/1

第一題

$V_T = 25mV$ $\quad V_{pb} = V_T \ln(\frac{N_A N_D}{{n_i}^2})$

(1) $N_A = 5 \times 10^{23}/cm^3$, $N_D = 3 \times 10^{19}/cm^3$

$$V_{pb} = 2.5 \times 10^{-3} \ln\left(\frac{15\times10^{42}}{2.25\times10^{20}}\right)$$

$$V_{pb} = 2.5 \times 10^{-3} \ln\left(\frac{20}{3} \times 10^{22}\right) = 2.5 \times 10^{-3} \ln\ (6.67 \times 10^{22})$$

第一題

$$V_T = 25mV \qquad V_{pb} = V_T \ln(\frac{N_A N_D}{{n_i}^2})$$

(2) $N_A = N_D = 8 \times 10^{18}/cm^3$

$$V_{pb} = 2.5 \times 10^{-3} \ln\left(\frac{64\times10^{36}}{2.25\times10^{20}}\right)$$

$$V_{pb} = 2.5 \times 10^{-3} \ln\left(\frac{256}{9}\times 10^{16}\right) = 2.5 \times 10^{-3} \ln\left(2.84 \times 10^{17}\right)$$

第二題

- 截面積 $A = 300mm^2 = 3cm^2$
- 長度 $L = 2cm$
- 參雜濃度 $N_A = 5 \times 10^{15}/cm^3$
- 外加電壓 10V

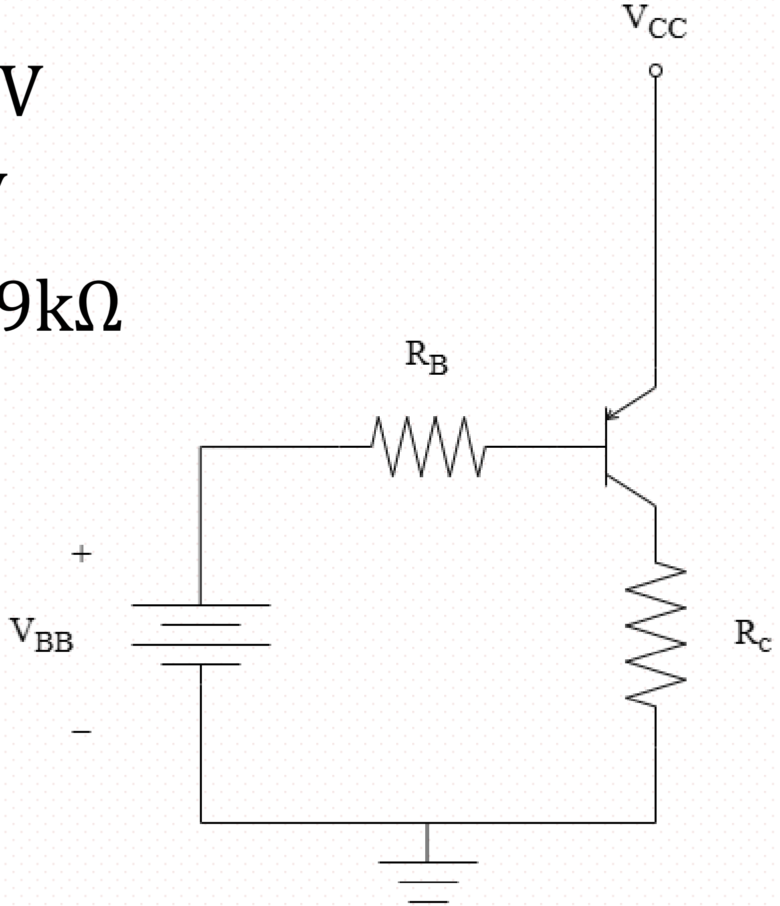
$$\rho = \frac{1}{q\,N_A\,\mu_P} = \frac{1}{1.6 \times 10^{-19} \cdot 5 \times 10^{15} \cdot 450} = \frac{1}{0.36}\,(\Omega \cdot \mathrm{cm})$$

$$R = \rho \cdot \frac{L}{A} = \frac{1}{0.36} \cdot \frac{2}{3} = \frac{50}{27}(\Omega) = 1.85(\Omega)$$

$$I = \frac{V}{R} = \frac{10}{\frac{50}{27}} = \frac{27}{5}(A) = 5.40(A)$$

第三題

- $V_{CC} = 20V$
- $V_{BB} = 6V$
- $R_B = 15.9k\Omega$



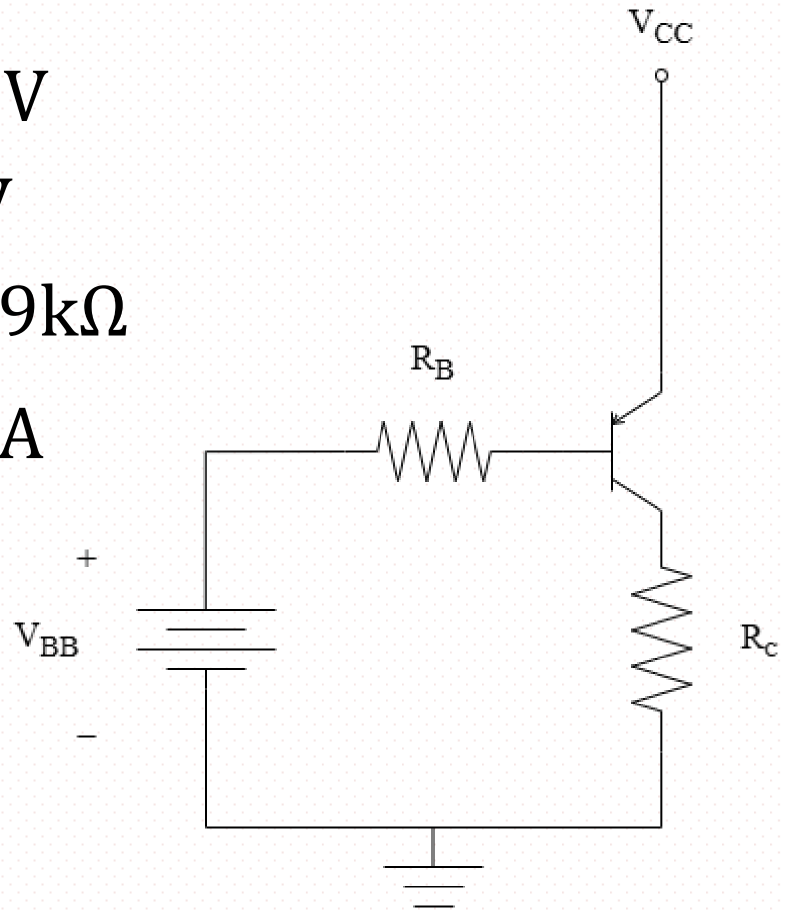
$$V_{CC} - V_{EB(ON)} - I_B R_B - V_{BB} = 0$$

$$20 \;-\; 0.7 \;-\; 15.9 I_B - 14 \;=\; 0$$

$$I_B = \frac{5.3}{15.9} = \frac{1}{3}(mA)$$

第三題

- $V_{CC} = 20V$
- $V_{BB} = 6V$
- $R_B = 15.9k\Omega$
- $I_B = \frac{1}{3}mA$



(1) $R_C = 6k\Omega$

假設主動模式(act)：

$$I_C = \beta I_B$$

$$I_C = 100 \times \frac{1}{3} = \frac{100}{3}(mA)$$

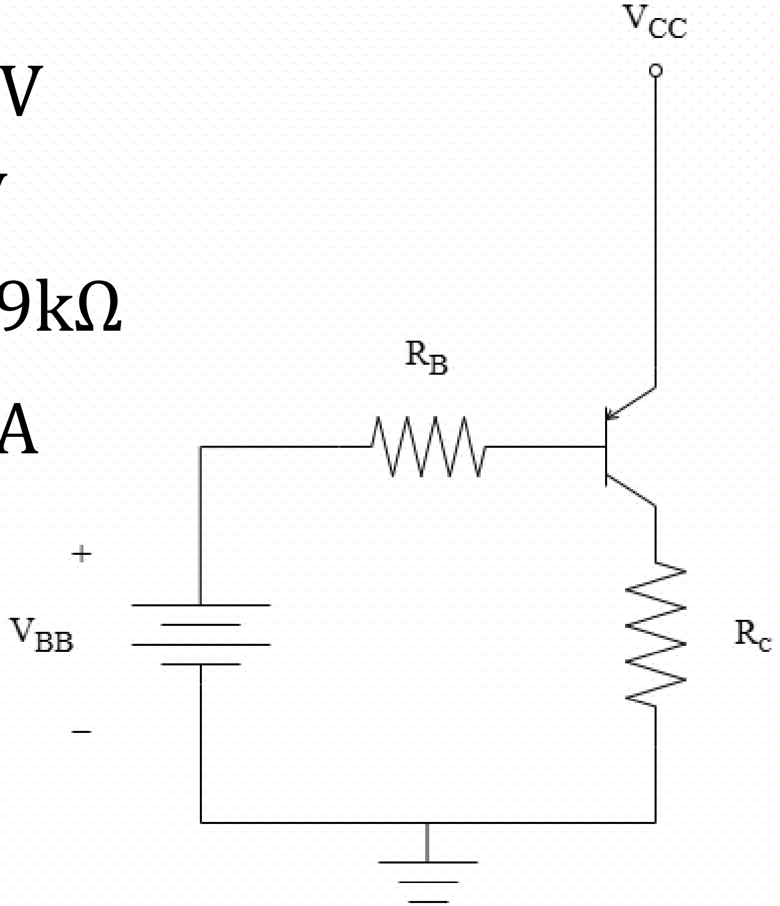
$$V_C = I_C R_C = \frac{100}{3} \times 6 = 200(mA)$$

$$V_E = V_{CC} = 20V$$

$$V_{EC} = 20 - 200 = -180 < 0.3 \ \dots (\times)$$

第三題

- $V_{CC} = 20V$
- $V_{BB} = 6V$
- $R_B = 15.9k\Omega$
- $I_B = \frac{1}{3}mA$



(1) $R_C = 6k\Omega$

假設飽和模式(sat)：

$$I_C = \frac{V_{CC} - V_{EC(SAT)}}{R_C} = \frac{20 - 0.2}{6} = 3.3(mA)$$

(Check ：$I_C < \beta\ I_B$)

$$I_E = I_C + I_B$$

$$I_E = 3.3 + \frac{1}{3} = 3.63(mA)$$

Ans：
$I_C = 3.3mA$
$I_E = 3.63mA$

第三題

假設主動模式(act)：

$I_C = \beta I_B$

$I_C = 100 \times \frac{1}{3} = \frac{100}{3}(mA)$

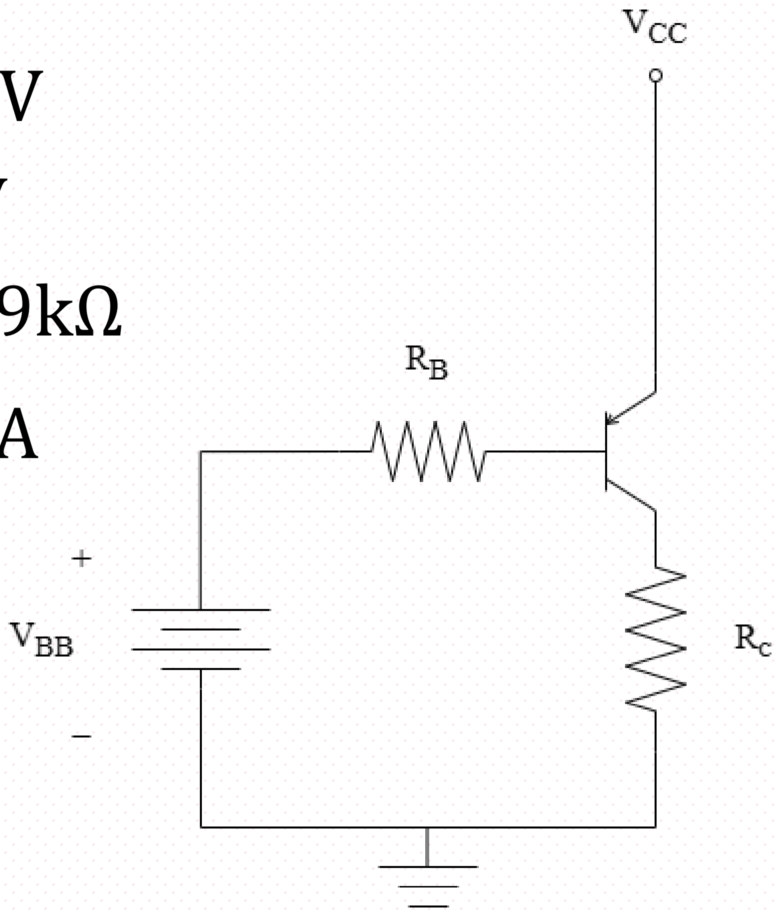
$V_C = I_C R_C = \frac{100}{3} \times 0.08 = 2.66(mA)$

$V_E = V_{CC} = 20V$

$V_{EC} = 20 - 2.66 = 17.33 > 0.3 \ \ldots (\checkmark)$

$I_E = (\beta + 1)\ I_B = 33.67(mA)$

- $V_{CC} = 20V$
- $V_{BB} = 6V$
- $R_B = 15.9k\Omega$
- $I_B = \frac{1}{3}mA$



(2) $R_C = 80\Omega$

Ans：

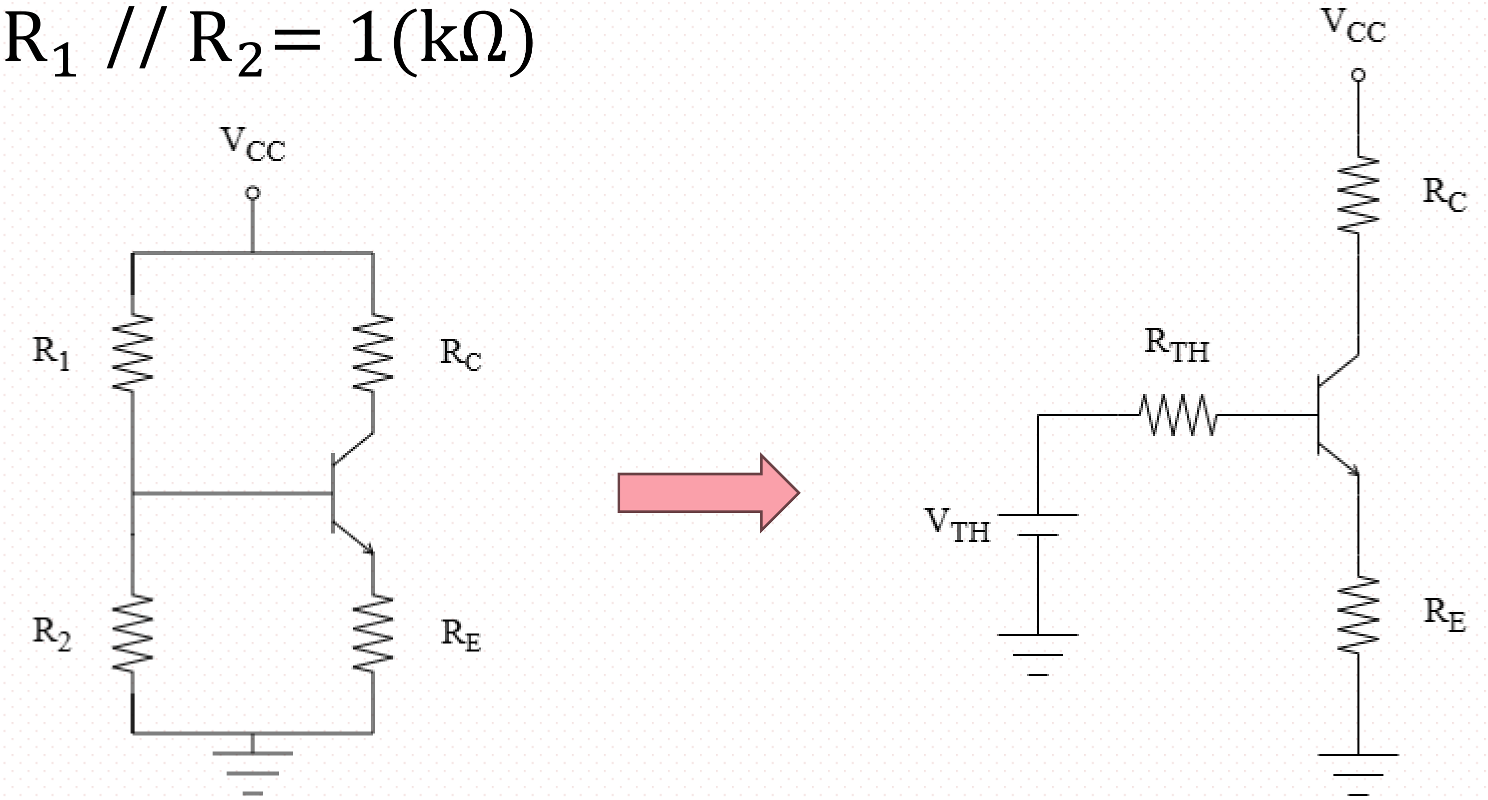
$I_C = \frac{100}{3}mA$

$I_E = 33.67mA$

第四題

$$V_{TH} = V_{CC} \times \frac{R_2}{R_1 + R_2} = 42 \times \frac{2}{2+2} = 21V$$

$$R_{TH} = R_1 \ // \ R_2 = 1(k\Omega)$$



第四題

假設主動模式(act)：

$$V_{TH} - I_B R_{TH} - V_{BE(ON)} - I_E R_E = 0$$

$$21 - I_B - 0.7 - (101 \cdot I_B \cdot 2) = 0$$

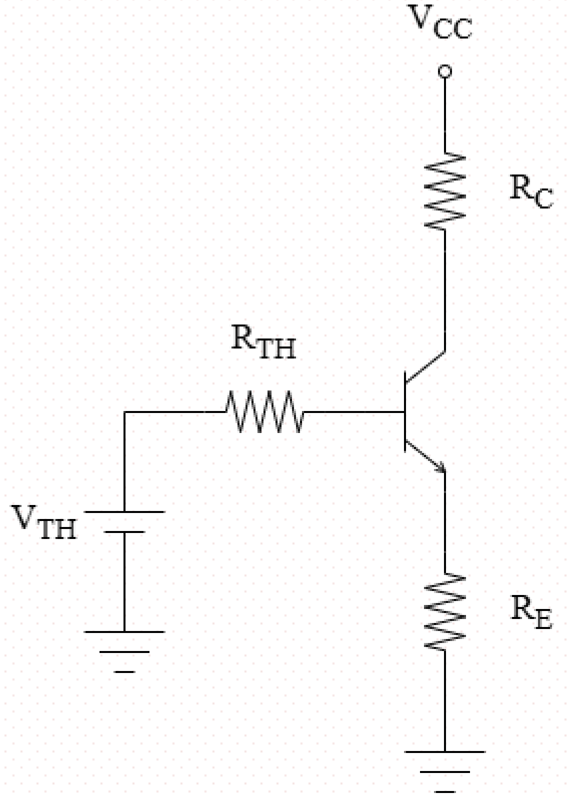
$$20.3 = I_B \cdot 203$$

$$I_B = 0.1(\text{mA})$$

$$I_C = \beta I_B = 100 \times 0.1 = 10(\text{mA})$$

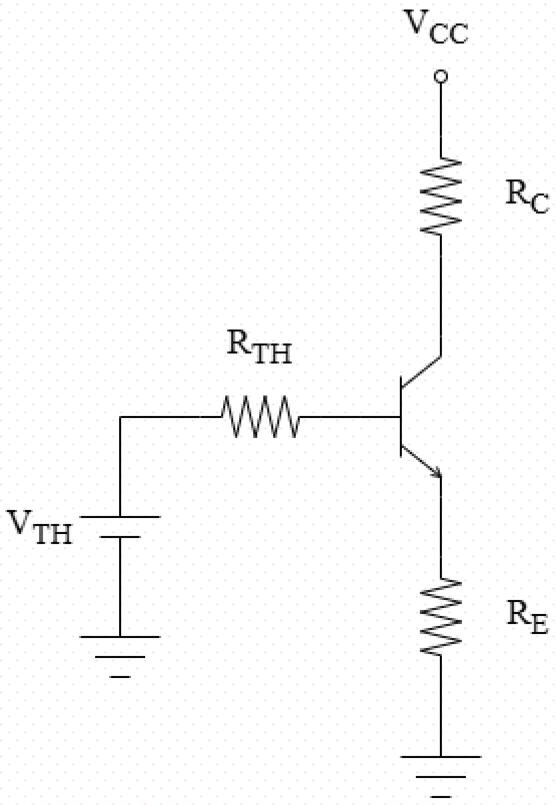
$$I_E = I_C + I_B = 10 + 0.1 = 10.1(\text{mA})$$

- $V_{TH} = 21\text{V}$
- $R_{TH} = 1\text{k}\Omega$
- $R_C = 1\text{k}\Omega$
- $R_E = 2\text{k}\Omega$



第四題

- $V_{TH} = 21V$
- $R_{TH} = 1k\Omega$
- $R_C = 1k\Omega$
- $R_E = 2k\Omega$



$$V_E = I_E R_E = 10.1 \times 2 = 20.2V$$

$$V_C = V_{CC} - I_C R_C = 42 - (10 \times 1) = 32V$$

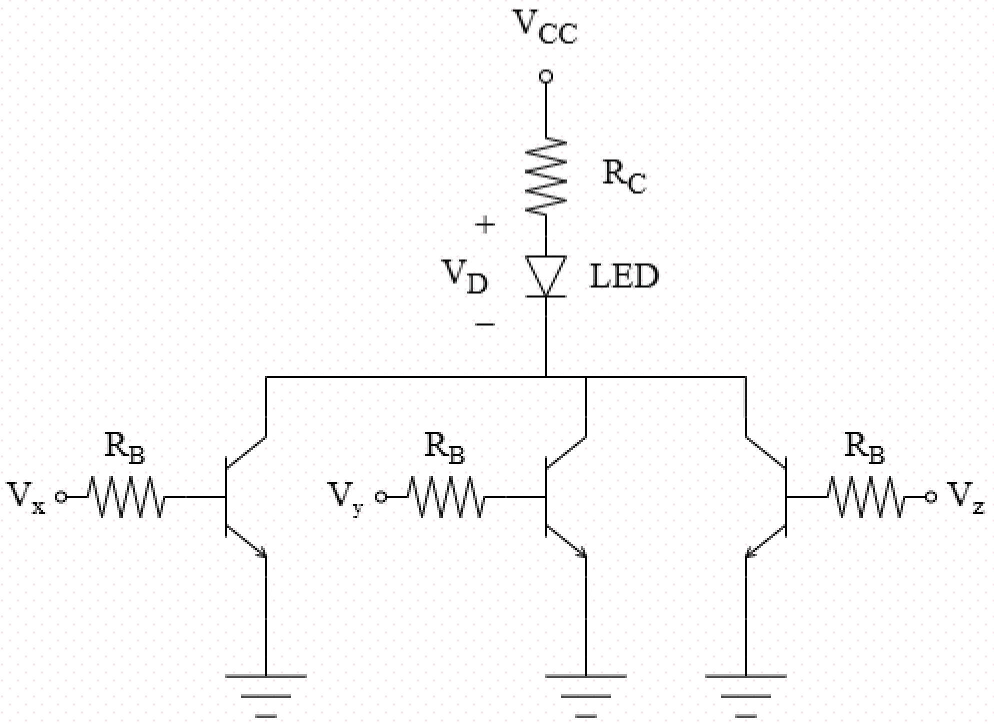
(check)

$$V_{CE} = V_C - V_E = 32 - 20.2 = 11.8 > 0.3 \ldots (\checkmark)$$

第四題_Ans

I_B	I_C	I_E	V_C	V_E
0.1mA	10mA	10.1mA	32V	20.2V

第五題



$P = SI$

$300 = 5 \times I_C$

$I_C = 60(\text{mA})$

V_x or V_y or $V_z = 16V$ 時，至少一 BJT 在飽和模式

$V_{CC} - I_C R_C - V_D - V_{CE(SAT)} = 0$

$8 - (60 \cdot R_C) - 3 - 0.2 = 0$

$R_C = 0.08(\text{k}\Omega) = 80(\Omega)$

第五題

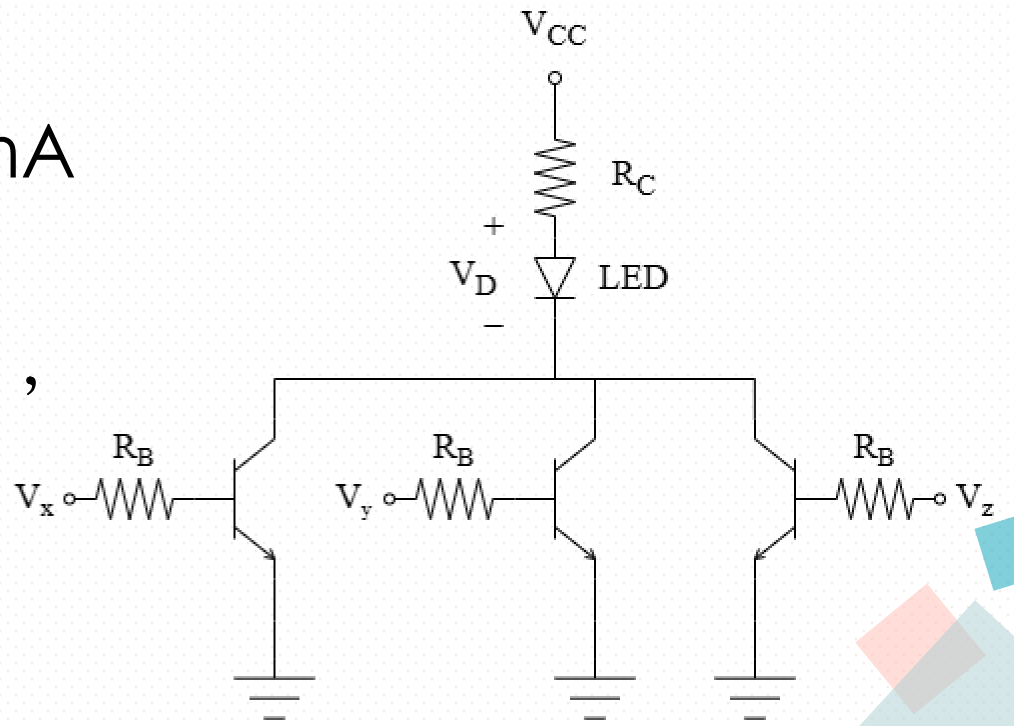
題目 BJT 為並聯，LED 導通的條件是至少一個輸入為 16V

最佳情況：通三顆，每顆 BJT 只需 20mA

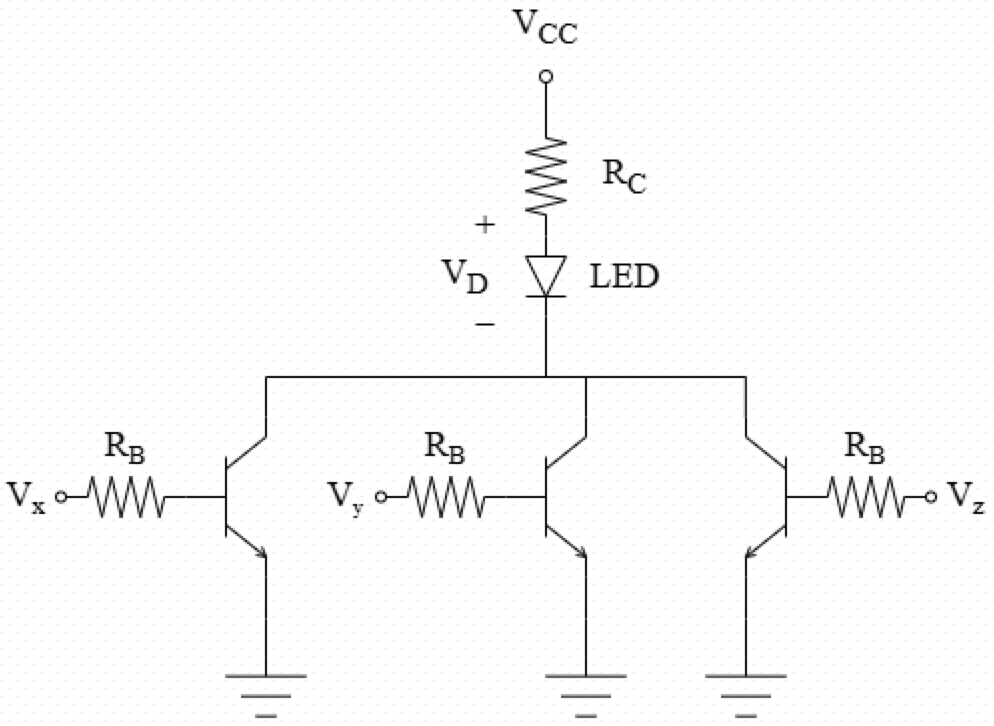
最差情況：通一顆，單獨一顆 BJT 需 60mA

要保證在最差情況下依然能進入飽和模式，

因此必須用 60mA 來推算所需的最小I_B



第五題



$$I_C \le \beta I_B$$

$$I_C \le \beta \frac{V_X - V_{BE(ON)}}{R_B}$$

$$60 \le 100 \cdot \frac{16 - 0.7}{R_B}$$

$$0.6 \le \frac{15.3}{R_B}$$

$$R_B \le 25.5(k\Omega)$$

$$R_B = 25\ (k\Omega)$$